Fish SRS Doc

4.1 Database Requirements

1. Introduction

1.1 Purpose

This application is a video game, meant for entertainment. Using Godot to create a 3d video game that is themed around fishing that is also a horror game.

1.2 Intended Audience

Other software developers that would like to recreate or improve the build for the game. It can also be used as an example of what to do and not to do.

2. Overall Description

2.1 Game Concept

Trash island, surrounded by spires and spikes that look like claws/fangs reach up into the sky. The island itself should be a mixture of sand and garbage, but mainly garbage. Dark stormy clouds with drizzles sometime, purple sludge and crystals everywhere for dim lighting. Shipwreck on a trash island, there was another with you but as you approached the unconscious body. A fish a medium size of a dog, pounce out of the water fins leg like as it bites and drag the man into the water. Starting from the pain, the man startles and screams in agony as he is dragged down below. It would be a good idea to not be so close to the water, or at least a beach where fish can come up and bite you. You traverse a little further inland, finding traces of others that might exist here on the island. You end up on a cliff edge, with a hut, a skeleton, and a journal. A grim warning, the only way to survive is to fish, but fish are mutated and they will fight back.

Gameplay - Play Loop

I really like how Monster hunter wilds did their fishing minigame. Walking around in a third person view, but then the camera can focus in for fishing. Fishing will be the main focus on the zoom in mode, where the player model becomes the background. With the purpose of making the game as realistic as possible, the player will just have a rod and a hook, not able to see the fish in the water at all. Fish, fight, loot, craft and repeat is the game loop I am going for. Since fish are mutated, and the theme is trash, fishing and slaying fish will give resources to upgrade gear, and some fish can also drop gear.

Gameplay - Setting what the environment would look like

Trash island, surrounded by spires and spikes that look like claws/fangs reach up into the sky. The island itself should be a mixture of sand and garbage, but mainly garbage. Dark stormy clouds with drizzles sometime, purple sludge and crystals everywhere for dim lighting.

2.2 Game Sequence

- Shipwreck
- Washed ashore
- Characters witness an npcs dragged into the water by a fish with jaws and fins forearms.
- Explore up the coast
- Finds hut cliff side
- Journal as a guide for new notes and unlock
- Eventually the idea to fish up trash god, slay it to actually be able to leave the island.

2.3 Software

For the Trash Game project, Godot and its coding language is selected. With experience working with 2d game dev with Godot, a 3d game would be the natural next thing I want to try. Godot has a 3d engine that flows really well if you have experience with its 2d game dev. Compared to other engines, Godot is meant for 2d games with a lighter asset library for 3d. Unreal and Unity are the other options, Unreal offering way too much of what I need and is looking for. Unity is beginner friendly and it has support for more targeted platforms. Godot is open source, and the price of free is hard to beat, plus I already have some experience with Godot and how it wants to organize itself.

2.4 Other Software

Blender

There is concern here for using blender, it has been stated that it has a tougher learning curve for beginners. With that said I have also messed around with Blender before, not to the extent I would like but better than nothing. Blender is also a free and open source option to use for 3d modeling.

The models and assets that I will be doing won't be high definition, which is one of the reasons why Unreal engine wasn't chosen. Graphics is not the purpose of this game, the graphics and aesthetic that I'm thinking of and looking for is closer to Peak, RV there yet, a grim dark fishing horror game with style and aesthetics as Pokopia.

Godot Terrain3D

Third party plug in that lets Godot model terrain needed for creation of trash island. Written in C++.

3. Milestones

3.1 Minimal Functional Requirements

What this project needs to achieve to be considered a minimal functioning.

1. Has a player model
    a. Human and 3d
2. Has the world map
    a. Starting shoreline
    b. Shack HQ
3. Fishing
    a. Equipment
        i. Rod
        ii. Reel
        iii. Lure
        iv. Line
    b. Fish
    c. Fish mechanic

For minimum functions, the user can control and walk around with the player model, go to a fishing spot to initiate the fishing game. There will be a shore, and there will be a hut where the user can edit equipment. The water will have at least one fish, for the user to fish the fish. Models needed, user, fish, equipment, a fishing rod would be four piece and require four different parts that would change how the rod behaves in game. Texture and looking pretty is not the priority, as this game is not meant for production, a demo at most.

3.2 Combat System

What does this project need for it to reach the next milestone?

1. Combat System
    a. User can attack and use skills
    b. Fish can attack and use skills
2. UI
    a. Player inventory
        i. Inventory/Backpack
        ii. Equipment slots
        iii. States
    b. Combat menu
        i. Attack
        ii. Skills
        iii. Run Up Shore
    c. Starting menu
        i. Resume or Load Save

ii. With save slots

iii. Options

iv. Exit

d. Overworld UI

The combat system is another push, where the game should be more developed and has the intended rpg turn base battle that is intended on its conception. All ui such as menus, inventory, and hud should be completed. The combat system must have a turn order that is established by speed stats of players and mobs. Players can attack, use a skill, or flee, mobs the fish will have random attacks and basic static ai for turn base fight.

3.3 Milestones

- [ ] 4/26/26
  - [ ] Models added
    - [x] ~~Player~~
    - [x] ~~Fish~~
    - [x] ~~Land~~
  - [ ] Equipment
    - [ ] Rod
    - [ ] Reel
    - [ ] Line
    - [ ] Hook

        - [ ] Water added
        - [ ] Fish into water
            - [ ] Behavior
            - [ ] Rarity
- [ ] 05/03/26
    - [ ] Gameplay
        - [ ] Cast
        - [ ] Reel
        - [ ] Fish behavior while reeling
        - [ ] Stat
    - [ ] Inventory
        - [ ] Fish resources and icon
- [ ] 05/10/2026
    - [ ] Minimal viable product
        - [ ] Player can walk around in the game
        - [ ] Player can approach and cast into water
        - [ ] Fish will give resources that will be added to inventory

4. Requirements

4.1 Database Requirements

No clue yet

Resources Used

SRS Doc

https://www.perforce.com/blog/alm/how-write-software-requirements-specification-srs-document#what-is-a-software-requirements-specification-document-01

Godot 3d Game demo

https://github.com/gdquest-demos/godot-4-3d-third-person-controller

Tempt player model

https://quaternius.itch.io/universal-animation-library

Godot set up instructions

https://quaternius.itch.io/universal-animation-library/download/eyJpZCI6MzQwODAzNCwiZXhwaXJlcyI6MTc3NjcwMzM2MH0%3d%2eS5YY%2bywb0KqRiz4fyh74BO4f1MA%3d

Tempt fish model

https://alstrainfinite.itch.io/fish

Terrain3D Documentation

https://terrain3d.readthedocs.io/en/stable/docs/introduction.html